

Anim Mirror Helper

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A tool to help the 3d animator in animating walks, runs and other looped animations.

What it does?

It creates a "cast" of the movement of one leg (or arm) to which another leg (arm) can be attached.

This cast moves the same way as the original leg, but in mirrored coordinates and with a specified time offset.

As a result, you animate only one leg, and the other leg makes exactly the same movement, but time-shifted.

Due to this, legs and arms are animated 3-6 times faster than the traditional way.

This is especially true for animal leg animations and the like with many controls and complex movements that are difficult to repeat and mirror on both legs.

Installing.

- Download the file, unzip the archive wherever you want and drag and drop the file
- **Drag_and_Drop_to_install.mel**
- onto your MAYA window.
- The tool icon will appear on the current shelf.

Using.

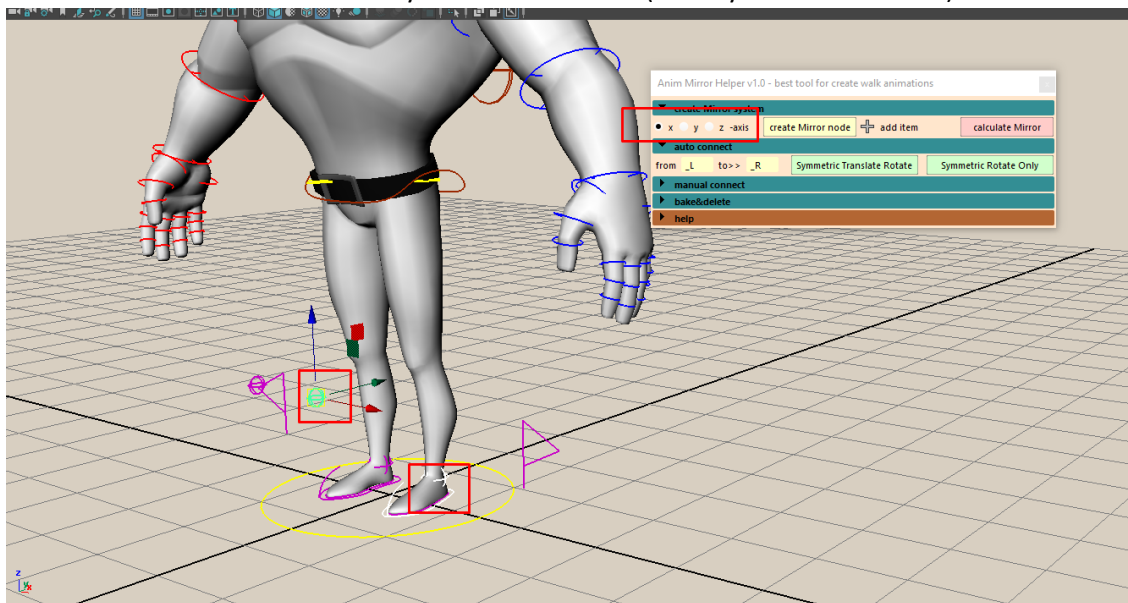
The general principle.

- You select the controls of one leg or arm that you want to mirror.
- You create a "cast" of the movement and a recalculation system.
- Connect the controls of the other leg to this mold.
- And by pressing the tool icon button, you recalculate the movement every time you change the animation.

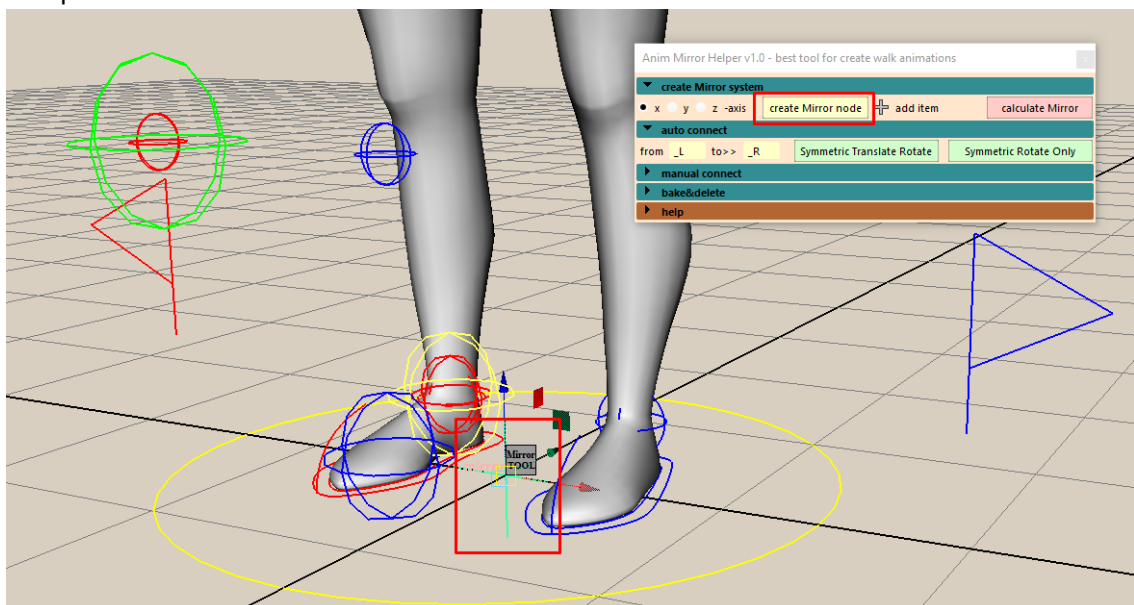
[In more detail with simple use](#)

Let's take the character's legs as an example (it's better to create different systems for arms and legs - it's more convenient)

- Start the tool.
- Select all the controls of one leg that you want to mirror.
- It is best to have the character standing in a mirror pose - it will be easier later on. But not necessarily.
- How to put the character in a mirror pose and not to break the animation, if it is already there, will be told later in the appropriate section.
- Choose the axis relative to which you want to mirror. (Usually this is the X-axis.)

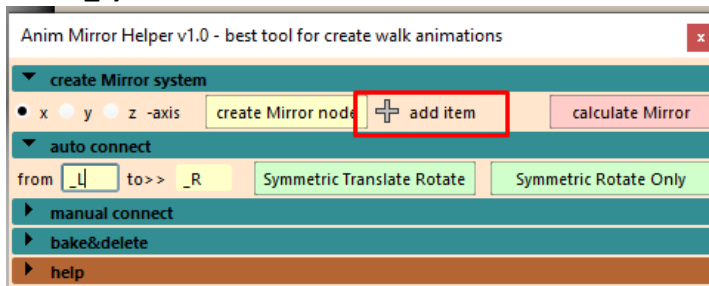


- And press the **Create MirrorNode** button

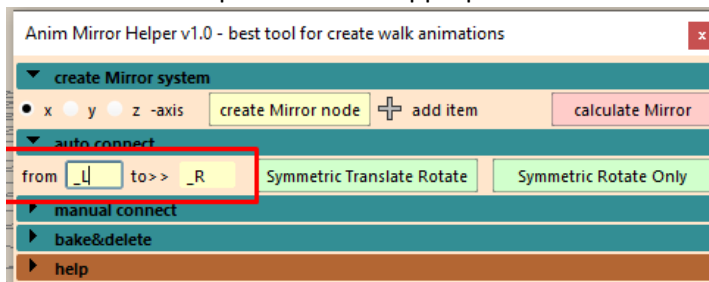


- A system of mirror objects will be created.
- And the object with the name will be selected **mirror_system**.
- You can move it to a convenient place for you

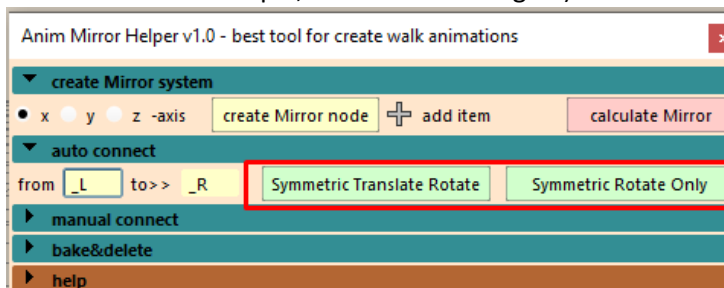
- If at this stage you forgot to add any objects - then select these objects, select **mirror_system** and click **Add Item**



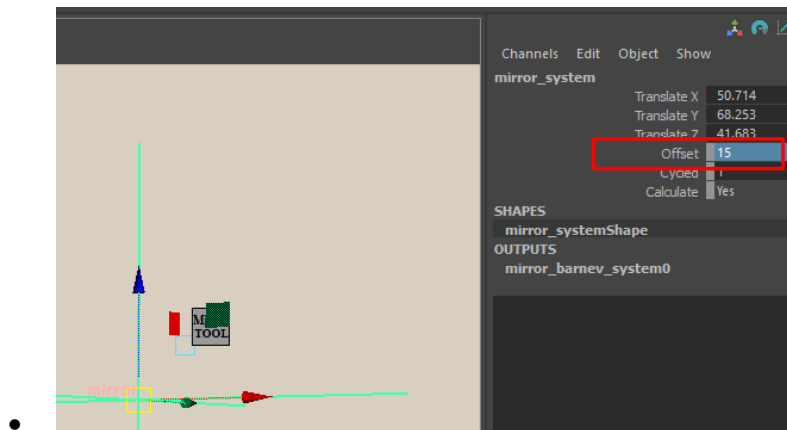
- After you have added all the necessary objects, you need to write the prefixes that denote the character's sides in this rig.
- Enter the desired prefixes in the appropriate fields.



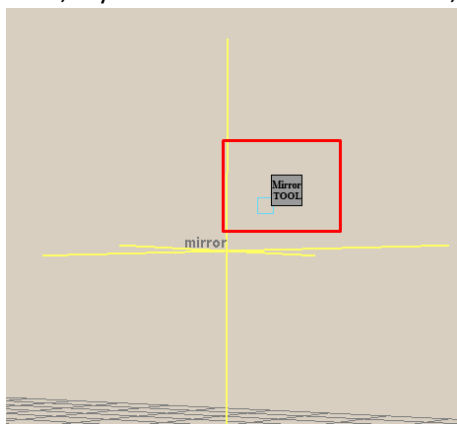
- After that, select the mirror_system object and press one of the two buttons **Symmetric Translate Rotate** (best for leg controls) or **Symmetric Rotate Only** (best for Forward Kinematics. For example, for hands and fingers)



- Then, if the character stands in a symmetric pose, the script will find the mirror names of the Rig controls and connect them to the mirror "casts".
- **Note that not only moves and rotations of objects are connected, but also values of attributes on controls!!!! I.e. you can mirror the whole leg movement system - IK/FK , animation via attributes and so on.**
- After that, select the **mirror_system** object and set the desired animation offset on the **Offset** channel. Usually, this value is equal to half of the animation cycle period



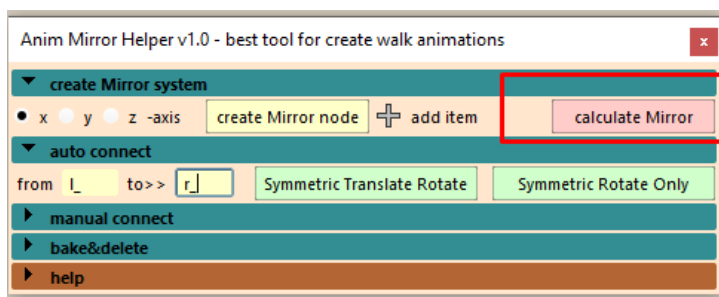
- Now, if you click the **Mirror Tool** icon, the leg animation is recalculated.



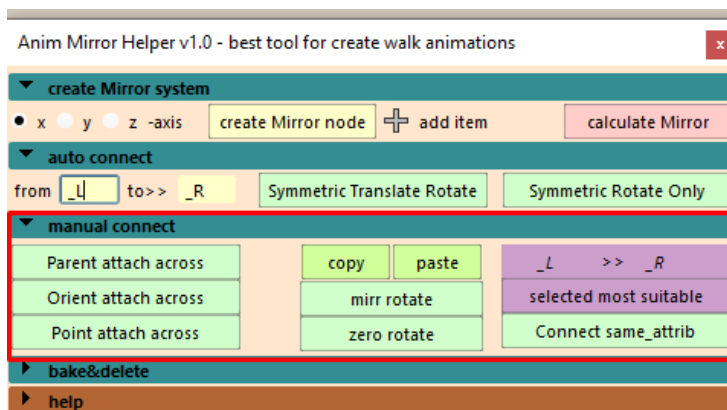
- And later, while you are animating one leg, you can press this icon, and the animation will be recalculated.
- You can do the same for the arms (arms are more convenient to do on a standalone system)
- Also, the arms can be rotated to a different angle - and the mirror animation will be transferred with this rotation. How to do this - see below. (for the legs, it almost does not make sense, although the cases may be different)

The **calculate Mirror** button is for when, for some reason, the button in the viewport does not work.

Then select the mirror_system object and click this button



MANUAL CONNECT tab



In my professional life as an animator, I've encountered a huge number of all sorts of rigs. And many of them are very STRANGE

For example, the orientation of the different parts - which may not be the same for the left/right halves.

Or their naming can be very strange.

Or you need to move the shift animation to another leg on the same side (e.g. from right front to right back).

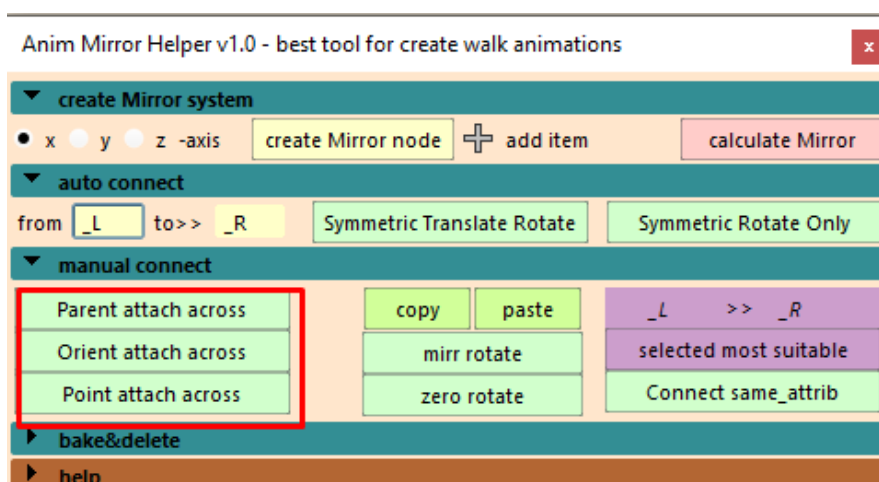
Or you want to fine tune the mirroring system.

And many other cases where you can't do it the standard way of connecting and have to do it manually.

That's what this tab is for.

How do you use them?

After you have made a system of mirror objects, you can connect the controls you need manually to them.



If you select a mirror object and a mirror control (you can choose several such pairs) and press one of these buttons, a locator will be created inside each mirror object and the corresponding control will be attached to this locator.

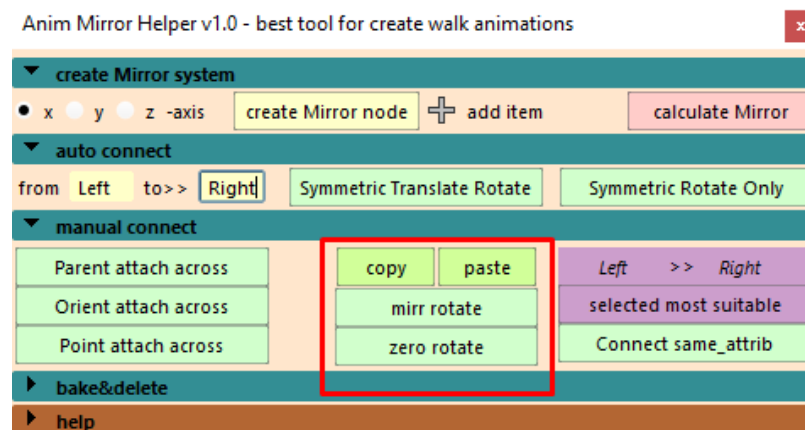
PARENT attach across - the control will be attached using **parent constraint** (i.e. it will repeat the movement of the source by translate and orient channels)

ORIENT attach across - the control will be attached using the **orient constraint** (i.e. it will repeat the source movement only through the orient channels)

POINT attach across - control will be attached with a **point constraint** (i.e. it will repeat the motion of the source by translate channels only)

After that, you need to make a relative rotation of the control relative to the mirror - by rotating these locators. This will be a different rotation for each rig. But usually it is either 0 0 0 or - (for the X axis) 180 0 0.

For this purpose the following column is made

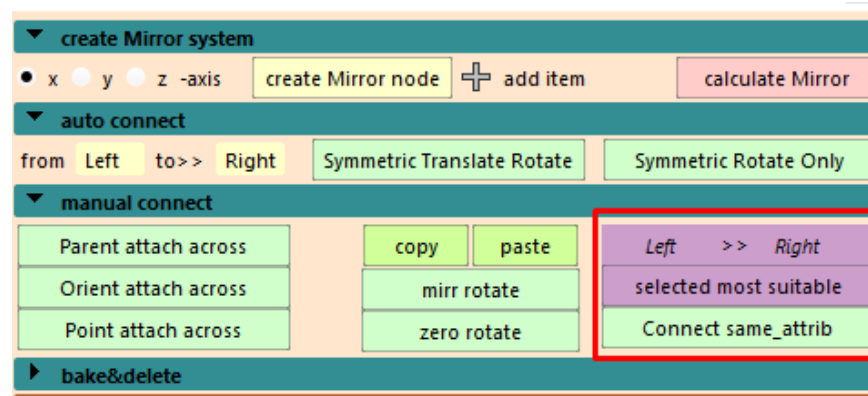


Copy\Paste - If a turn is non-standard, you can memorize it and then assign it to any selected internal locator.

Mirr Rotate - puts the rotate 180 0 0.

Zero Rotate - puts the rotate 0 0 0.

If the names of the rig objects are more or less standard, you can use the third column.



The first line simply repeats the names that are entered in the line **auto connect**.

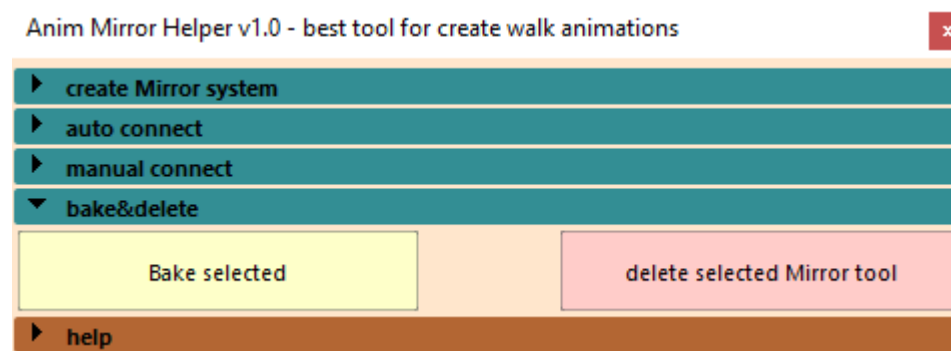
Selected most suitable - if you select mirror objects and press this button, the script will try to select the controls in pairs based on the specified prefixes (so that you can press the buttons from the first column at once) or to press the **Connect same_attr**

Connect same_attr - If you select a mirror object and a mirror control (you can select several such pairs) and press this button, then attributes with the same name will be docked between these pairs.

The point is that when you create mirror objects, they also have attributes taken from the original controls, with the same names. And they too will be shifted by a given number of frames.

If you have attributes on the left and right hand that do the same thing and have different names, you can connect them using the standard Connection Editor

Bake&Delete tab



After you have done the animation and are no longer going to change the legs, you can bake mirror controls.

They need to be selected (if you created them immediately - they lie in a specially created sets) or manually. And press the button **Bake Selected**.

The animation is baked onto these controls (with speed baking, which works up to 6 times faster than standard)

If you select **mirror_system** control and press **Delete Selected Mirror Tool** - the entire mirroring system will be deleted.

An important feature of this tool. Once you've set up mirroring, you can apply ANY animation to your original objects!

Manually, with animation layers, or with scripts that re-bake the animation.

In either case, the animation will reflect correctly on the mirrored controls.

More fine-tuning for better results. As well as tricks and tips for working with this tool

How to put a character in a mirror pose and not to break the animation, if it is already on it.

Often the animator gets the brilliant idea to mirror the animation - only in the middle of the work, when one leg is already animated.

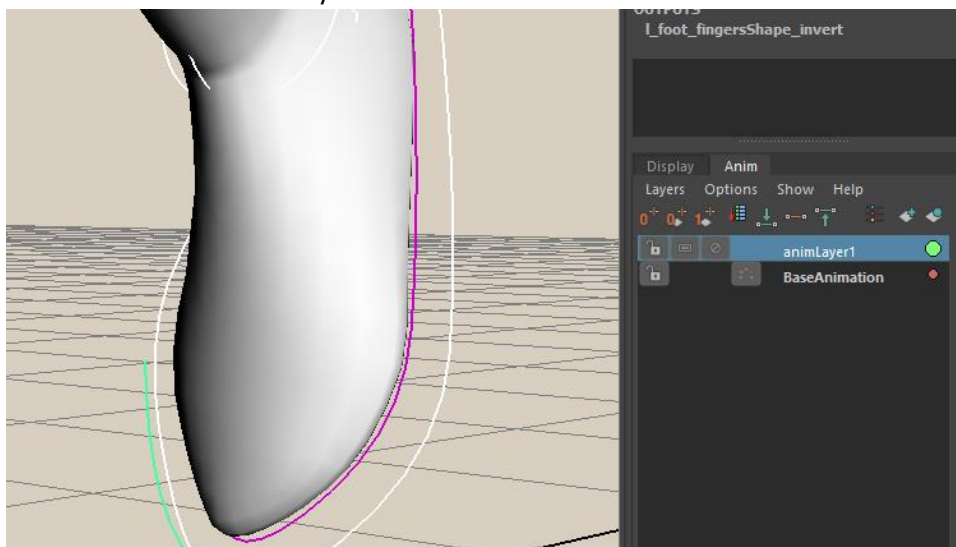
In principle, there is no problem to mirror and connect objects from any pose.

But it is more convenient when the legs (arms) are in a symmetric position.

What should we do in order to put the legs in a symmetrical position, and not to disturb the animation of the already animated leg?

Animation layers will help us!

- Select the controls of the leg you want to mirror.
- And make an animation layer with them.



- Set the keys
- And put the foot to the default position (usually it is enough to put the values 0 0 0 on translates and rotates)
- Then put the second leg in the same position in the same frame (you don't need to add it to the layer, because this leg will be controlled by the mirror system anyway)
- Now you can do everything else.
- After connecting the second leg to the mirror system you need to remove this animation layer.
- Voila!

The main principle of this tool

The group appears in the scene when it is created **mirror_system_gr**

Within it are the groups of **base_group_for_mirror** and **base_group_not_mirror**

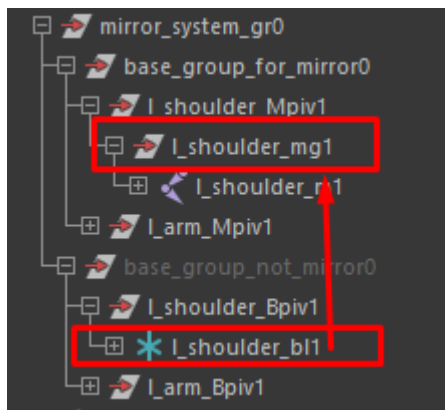
base_group_not_mirror - it contains the structures **from** which we transfer the animation

base_group_for_mirror - it contains the structures **that** we transfer the animation to

In turn, within these groups are the groups with the names of the objects we mirror.

At the same time, the transfer of animation is performed relative to these nested groups, i.e. in RELATIVE coordinates.

This is how (it is for one object):



Why is this important? And how can it be used?

For example, we want to mirror the movement of the arm and clavicle.

The arm is in FK mode.

By default, the arm and clavicle movements will be recalculated relative to the GLOBAL coordinates. That is, as long as the body motion is symmetrical in both directions, all will be fine.

But if, for example, the body is side tilted? Then the arm movement should still be relative to the global coordinates, but the clavicle movement should be recalculated relative to the chest. Then, no matter how the chest moves, no matter how asymmetrical it is, the motion of the clavicle will be recalculated correctly.

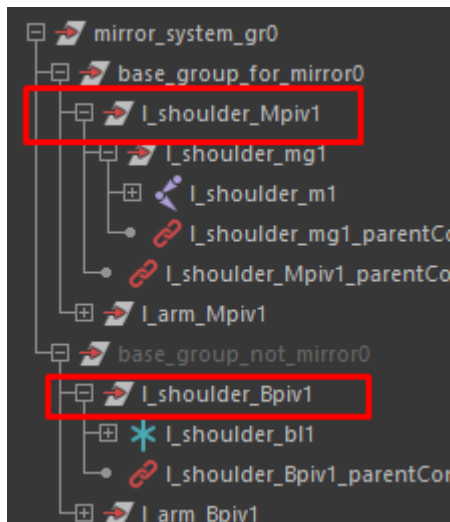
How to do it?

You need to make a parentConstraint with offsets to the chest control of these groups:

I_shoulder_Bpiv1 (top group of the object **from** which we take the animation)

And

I_shoulder_Mpiv1 (the upper group of the object, HERE we recalculate the animation)



If you do this, both of these groups will move and rotate together with the chest and, accordingly, will exchange RELATIVE motion coordinates.

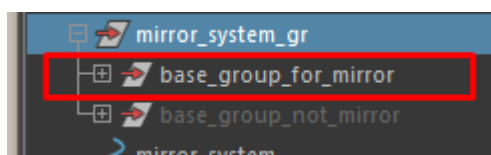
How to make a mirror movement, but at the same time rotated?

Very often you want to do this kind of arm movement

When one hand, for example, the left hand moves in one plane, and the right hand moves exactly the same way (with a shift of half a period), but rotated by some degree (for example, bringing the hand more strongly to the chest)

How do you do this?

Since the motion within the system is transmitted in relative coordinates, we can, for example, simply rotate **base_group_for_mirror** to the degree we want. Or even animate it, if necessary.



And then the hand will carry the mirror movement, but it will be rotated.

How to make a mirror movement, but I want to animate the object BEFORE the mirror animation, so that the system works?

For example. You want the arm to repeat the movement completely, but with a little more work on the forearm.

To do this, you can take a mirrored object that corresponds to the forearm and animate it already. These objects are specially made nested within the group to which the animation is recalculated, just for this purpose.

Can I use animation on the objects - sources with animation layers?

Yes. You can animate the original objects (arm, leg) ANY way you want.

With animation layers, or with animation scripts, etc.

Does having this system affect the performance of the Rig? Won't the scene slow down?

No. The presence of this system does not affect the performance of the scene in any way. It is specially made to calculate at the touch of a button. But if it were REALTIME, that would be a big problem on complex rigs.

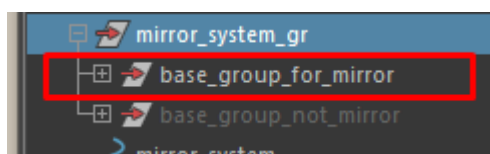
How do I make not a mirror, but just a movement with an offset in time?

For example, I want to transfer the movement not from the left to the right leg, but from the front to the back.

Is it possible to do that?

Yes, you can.

All you have to do is take **base_group_for_mirror**



And put it in the channelbox with the positive value of the skale .

And move it to the place relative to which the motion of the back leg will be recalculated.

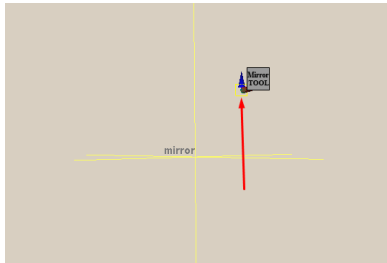
I'm animating a gait with a shift (the character moves forward instead of standing still)
how do I make the movement mirror correctly?

Since the recalculation is performed in relative coordinates, it is enough that **mirror_system_gr** moves together with the gait.

This can be done in two ways

1. Simply animate it - move it to the same distance as the general character shift per step.
2. If the character is shifted for a common MAIN CONTROL - just put it inside that control. Or constraintPoint it to it.

For ease of use - to easily hit the recalculation button - you can move it by this square relative to the control.



If I need to mirror object scales, what do I do?

To do this, you have to manually connect the corresponding controls to the mirror objects using `scaleConstraint`

I made a mirrored node and in the process of animation I realized that I forgot to add another object.

Yes, you can add another object at any time, just as described in the beginning. But you have to connect the opposite control to it manually.

I want to move the whole arm system - both IK and FK, how do I do that?.

Exactly the same as for the other objects. Switch to FK and add them to the mirror node. And also add a toggle object between them - then the channel animations of these toggles will be mirrored

If you have any questions, email me at pavel_barnev2000@mail.ru